Advanced database management systems

Topic 3: Data Quality

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Overview

- We measure the quality of data stored in various forms at a company
- We design and implement automated or interactive methods to improve the quality

After completing this topic:

- you will understand the hard and soft dimensions of data quality
- you will be able uncover hidden business rules in structured company data
- you will be able to design and run data cleansing and knowledge discovery projects using the MS DQS infrastructure

Types of business data

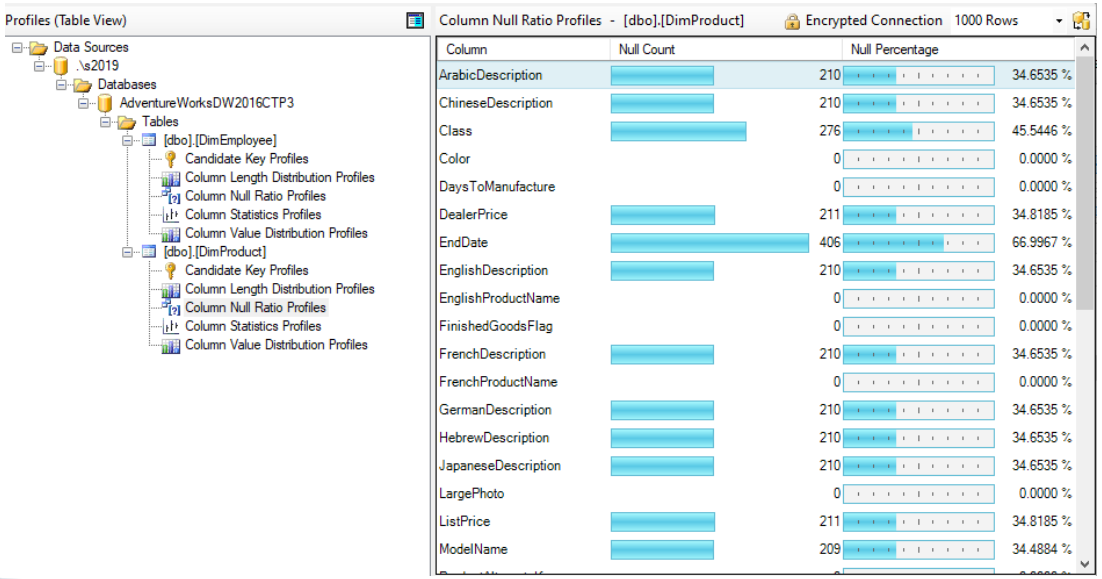
- OLTP, relational
- Data warehouses, star, hierarchical/multidimensional
- Semi structured, e.g. json
- Unstructured, e.g. emails
- Master data
- Metadata

Data quality dimensions

- Hard dimensions
  - Completeness—do we have all the data?
  - Accuracy—are the values accurate?
  - Consistency—do data contradict among different systems?
- Soft dimensions
  - From the users' perspective, e.g. trust.

Understanding data: Data profiling

- Column value distributions and other statistics
- Measure null ratio and length distribution for string types
- Enforced vs. *de facto* functional dependencies
  - Even if a dependency is not enforced by the domain modell, it may be required *de facto*
  - Violating hidden dependencies may lead to **redundancy and inconsistency**



SS DQS Data Cleansing projects

- Kowledge base: a collection of domains
  - Domain name and base type, e.g. string
  - Reference data: domain values
    Leading value + synonyms
  - Normalization procedure e.g. to_lower()
  - Domain rules to be enforced for all valid values
    - May use a column pattern detected by Data Profiling
  - Term based relations
- Project configuration
  - Min score for suggestions, default 0.7
  - Min score for auto correction , default 0.8
- Results: Correct/Corrected/Suggested/New/Invalid

Building a Knowledge Base

- Knowledge discovery with a „gold standard” sample
- Manual domain preparation
  - Adding rules
  - Setting Errors and Synonyms
- Publishing and sharing the Knowledge Base